



Voraxia Ore System Reference

Data Assets, Gameplay Tags, Registry Lookup, Material Mapping, and Prototype Balancing

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VORAXIA

1. What This System Is For

The ore system connects editable voxel asteroids to the rest of Voraxia: mining yields, player inventories, exchanges, construction, scanner readouts, visuals, and later contracts or manufacturing. The goal is that an asteroid is more than a shape with a material applied to it - it contains mapped resources that can be revealed and vaporised by the Raptor.

Core rule: Gameplay Tags identify what an ore is; Ore Definition Data Assets define how it behaves and looks; the Ore Registry resolves a tag into its definition.

Voxel cell -> Resource.Ore.Copper -> DA_Ore_Copper -> material, yield, value, resistance, icon, debug colour

2. System Architecture

Gameplay Tag	Stable identity used in code, voxel cells, inventory, crafting, scanners, exchange records, contracts, and networking.
Ore Definition Data Asset	Per-ore configuration: display name, description, surface material, icon, debug colour, yield, resistance, value, and generation rules.
Ore Registry Data Asset	Central lookup table that maps a gameplay tag to the appropriate Ore Definition asset.
Voxel Resource Grid	Runtime asteroid data storing which ore occupies each voxel cell and how much remains to be mined.

3. Source Layout and Asset Layout

Source/VoraxiaVoxel/Public/Resources/VoraxiaOreTags.h

Source/VoraxiaVoxel/Private/Resources/VoraxiaOreTags.cpp

Source/VoraxiaVoxel/Public/Resources/VoraxiaOreDefinition.h

Source/VoraxiaVoxel/Public/Resources/VoraxiaOreRegistry.h

Source/VoraxiaVoxel/Private/Resources/VoraxiaOreRegistry.cpp

Recommended Content Browser folder:

Content/_Voraxia/Resources/Ores/

Recommended data assets:

DA_Ore_Rock, DA_Ore_Iron, DA_Ore_Copper, DA_Ore_Cobalt, DA_Ore_Nickel, DA_Ore_Ice, DA_Ore_Titanium, DA_Ore_Gold, DA_Ore_Silicon, DA_Ore_VolatileCrystal, and DA_OreRegistry.

4. Ore Catalogue

The early prototype uses clear, legible materials that can be recognised in a quick glance. Start with a small subset in live asteroid generation, then expand after the yield and inventory loop is verified.

Ore	Gameplay Tag	Yield / Cell	Resistance	Base Value	Visual	Data Asset
Rock	Resource.Ore.Rock	0.10	0.75	1	Dark grey	DA_Ore_Rock
Iron	Resource.Ore.Iron	0.50	1.00	8	Steel grey	DA_Ore_Iron
Copper	Resource.Ore.Copper	0.65	0.85	12	Orange	DA_Ore_Copper
Cobalt	Resource.Ore.Cobalt	0.60	1.15	20	Deep blue	DA_Ore_Cobalt
Nickel	Resource.Ore.Nickel	0.75	1.10	18	Pale green silver	DA_Ore_Nickel
Ice	Resource.Ore.Ice	1.50	0.60	5	Cyan	DA_Ore_Ice
Titanium	Resource.Ore.Titanium	0.35	1.75	45	Pale silver	DA_Ore_Titanium
Gold	Resource.Ore.Gold	0.15	1.25	80	Warm gold	DA_Ore_Gold
Silicon	Resource.Ore.Silicon	0.45	0.95	16	Black crystalline	DA_Ore_Silicon
Volatile Crystal	Resource.Ore.VolatileCrystal	0.05	2.00	120	Purple	DA_Ore_VolatileCrystal

Balancing note: yield values are intentionally fractional. Multiplayer inventories should ultimately store a fixed-point smallest unit, for example 1,000 internal units = 1.0 displayed ore unit, rather than raw float values.

5. Visual Reference Icons

These generated icon references are suitable as directional art for the Data Asset inventory icon field. They are not required for the ore system to function.



6. Ore Definition Data Asset

Each ore receives its own UVoraxiaOreDefinition Data Asset. The Gameplay Tag is the stable identity; this asset is where designers tune the actual behaviour and presentation.

Property	Type	Purpose
OreTag	FGameplayTag	The permanent identifier, such as Resource.Ore.Copper. Use this everywhere instead of checking a material asset directly.
DisplayName	FText	Player-facing label for UI, scanner readouts, exchanges, inventory, and tooltips.
Description	FText	Short player-facing explanation of the material and its industrial role.
AsteroidSurfaceMaterial	UMaterialInterface	The material applied to exposed ore veins or pockets. For the MVP this can be a flat-colour Material Instance.
DebugColour	FLinearColor	Used by debug boxes, scanner pings, temporary UI, and early ore visualisation.
InventoryIcon	UTexture2D	The icon used by player inventory, ship inventory, exchange UI, and craft screens.
YieldPerVoxelCell	float	The amount awarded once a resource voxel cell is fully vaporised. Must allow values below 1.0.
MiningResistance	float	Controls how hard the material is to cut. 1.0 is normal, lower is easier, higher is tougher.
BaseExchangeValue	int32	Prototype baseline value per displayed ore unit at an exchange.
bCanAppearOnSurface	bool	Whether deposits may intersect the asteroid surface and create visible veins or pockets.

7. Important Yield Change

The original first-pass code used an integer yield with ClampMin = 1. That blocked values below one. The correct prototype direction is a float yield property with a minimum of 0.01 so common material and trace deposits can grant small fractions.

```
UPROPERTY(EditDefaultsOnly, BlueprintReadOnly, Category = "Voraxia | Ore | Mining",
meta = (ClampMin = "0.01", UIMin = "0.01", UIMax = "10.0"))
float YieldPerVoxelCell = 1.0f;
```

8. Registry Lookup

DA_OreRegistry is the central catalogue of ore definitions. It contains the set of DA_Ore_* assets and resolves a gameplay tag into its definition at runtime.

- 1. Voxel cell stores a tag** Example: Resource.Ore.Copper
- 2. Asteroid asks the Ore Registry** FindOreDefinition(Resource.Ore.Copper)
- 3. Registry resolves DA_Ore_Copper** Returns material, yield, value, colour, resistance, and icon.
- 4. Mining applies result** Raptor vaporises the cell; the asteroid awards the configured amount exactly once.

9. Runtime Mining Flow

Client requests Raptor action

- > Server validates hit and range
- > Asteroid identifies affected resource cells before density is removed
- > Density carve runs
- > Fully vaporised cells yield their resource amount once
- > Asteroid updates resource remaining state and remeshes
- > Server updates inventory and replicates the result

The asteroid owns resource evaluation. The Raptor does not decide that it mined gold, copper, or ice. This keeps handheld tools, ship drills, cutters, weapons, and salvage systems consistent and protects authoritative multiplayer state.

10. Asteroid Generation Rules

- Each asteroid should contain no more than three ore types.
- Rock fills all cells not claimed by an ore deposit.
- Deposits can be internal pockets, stretched veins, or surface-biased deposits.
- Surface ore is simply a deposit that intersects the outer density boundary.
- Deep ore is revealed by carving toward a hidden deposit or later by scanning.
- For the MVP, begin with Rock, Iron, Copper, and optionally Ice. Keep Cobalt, Nickel, Titanium, Gold, Silicon, and Volatile Crystal registered but unused until needed.
- Final authored asteroid meshes should be voxelised offline into density and material/resource data, while the current voxel sphere remains a test platform.

11. Six-Month Reminder Checklist

- Gameplay Tags identify ore types. Do not use materials as ore identity.
- DA_Ore_* assets hold visuals, values, yield, resistance, and icon data.
- DA_OreRegistry is the one central tag-to-definition lookup asset.
- Voxel cells should store lightweight resource identity and remaining yield data.
- Each cell must only award yield once when fully vaporised.

- YieldPerVoxelCell is float and supports fractional values; player/ship inventories should eventually use fixed-point integers internally.
- Keep asteroid contents to three ore types maximum for readability and balancing.
- Multiplayer server owns voxel edits, yield calculation, inventory awards, and persistence.
- MVP order: resource grid -> test deposits -> debug visuals -> log yields -> player inventory -> final mesh material blending.

12. What To Build Next

A. Resource-aware voxel sphere	Add cell-based ResourceTypes and ResourceRemaining grids to AVoraxiaVoxelSphereActor.
B. Two test deposits	Place Iron and Copper blobs in the sphere; use debug boxes or colours to verify their positions.
C. Mining yield logging	When the Raptor removes a cell, log the actual tag and yield, and prevent duplicate grants.
D. Inventory handoff	Return an FVoraxiaMiningYield to the mining component, then insert it into player inventory.
E. Visual exposure	Pass resource data through Marching Cubes so newly exposed faces render with the correct ore vein material.

Voraxia ore-system principle: when the Raptor vaporises material, the asteroid knows what was removed, the server awards it once, and the player can see why that material matters.